

Technical Datasheet of WIN-IO-8UI8UOM-E Module (Modbus TCP/IP)

1. Overview

The Universal Input (UI) and Universal Output (UO) board support **communication exclusively over Modbus TCP/IP**, enabling seamless control and monitoring of inputs/outputs. The board provide **configurable input and output selection via TCP/IP web pages**, ensuring flexibility across various applications. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.

Digital IO module is sturdy, low power usage and easy to use.



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

Specifications

<u>Category</u>	<u>UIUO Module</u>
Input Type	Analog Input (10V) Analog Input (5V), Digital Input , NTC Sensor Input. (Configurable via TCP/IP web page)
Output Types	Analog Output (AO), Digital Output (DO). (configurable via TCP/IP web pages)
Connectors	Single port handling multiple input/output types

General –

Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 15 – 40 V DC or 24 V AC/DC Typical – 24V DC @ 150mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



UI (Universal Inputs) –

Digital Channels	8
Sense Voltage	3.3 – 12 VDC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec
Analog Channels	8 (Can sense Analog Voltage, NTC)
Input Signal	0 – 5VDC / 0 – 10VDC / NTC (Web page selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 15 VDC min
AI input impedance	120Ω

UO (Universal Outputs) –

Channels	8
Analog Output Resolution	10 bits (0 to 10VDC)
Channels	8
Digital Output	NPN Transistor based switching

Additional Features: -

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, ± 30 KV contact, ± 30 KV air

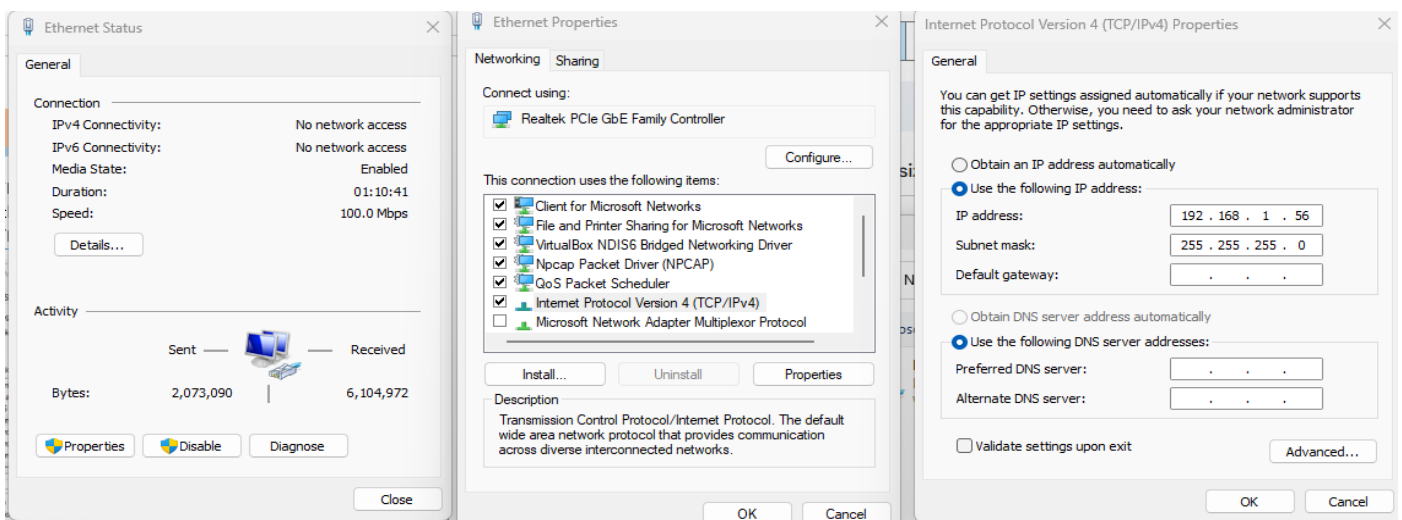
Configuration Of UIUO module

The UIUO module can be configured as Analog Input, NTC Sensor Input or Digital Input for each of the 8 UI. Similarly each of the 8 UO ports can be configured as Analog Output or Digital Output.

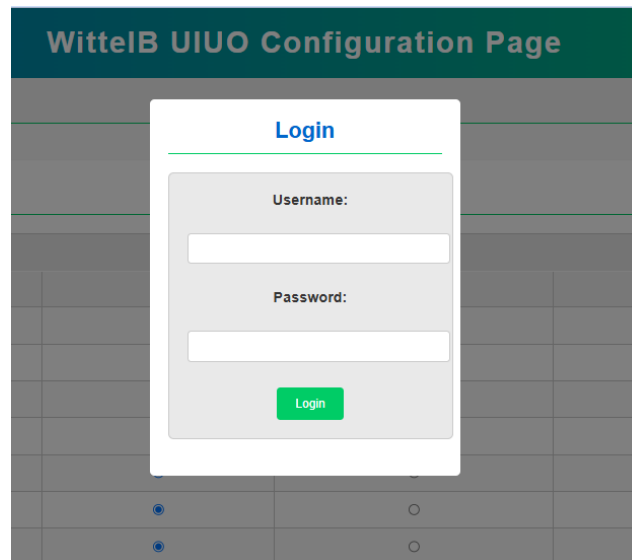
All the configuration can be done through Web page hosted at the default IP address (i.e. 192.168.1.177) or at the changed IP address.

The following are step by step guide for configuration:

1. Connect the Module to +24VDC POWER SUPPLY.
2. Connect the Board to the PC via an Ethernet Cable.
3. Use the default setting for PC as TCP/IPv4 (i.e., Address -192.168.1.56).



4. Now, open web browser and type **192.168.1.177/** (default IP address) and Press 'Enter'. A login page will appear as shown below.



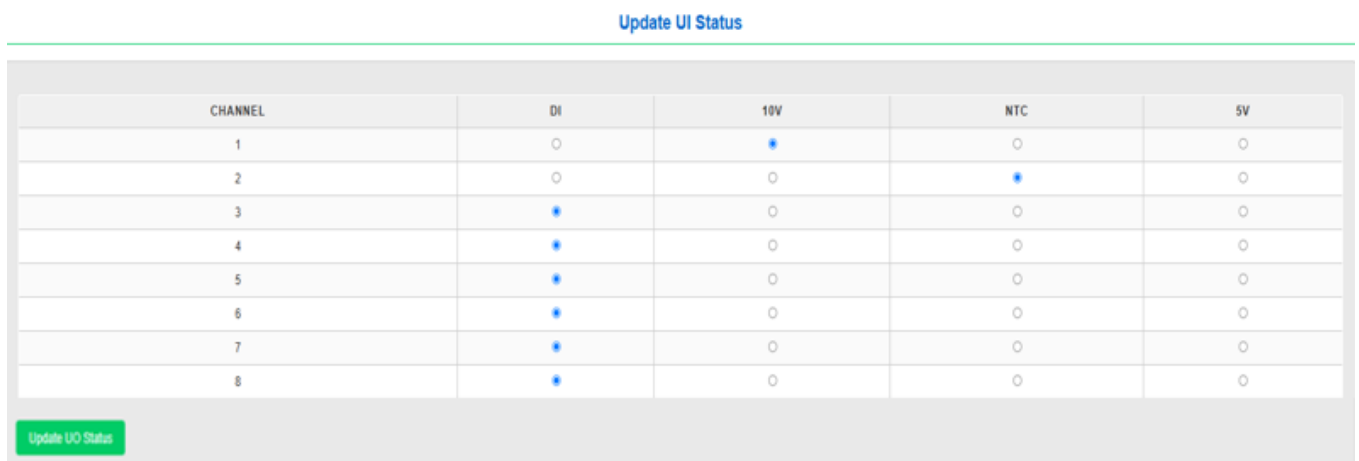
The image shows a web browser window displaying the 'WittelB UIUO Configuration Page'. In the center, there is a white 'Login' form. The form has a title 'Login' in blue. Below the title, there are two input fields: 'Username:' and 'Password:'. At the bottom of the form is a green 'Login' button. The background of the page is a dark grey grid pattern.

5. Default Login Credentials are :

Username: admin

Password: admin

6. After successful login, The UIUO Configuration page will be visible. It has three tables. The first and second table named as 'Update UI Status' and 'Update UO Status' which have selectable radio button for doing the configuration of respective channels as shown below.



The image shows the 'Update UI Status' configuration page. It features a table with 5 columns: CHANNEL, DI, 10V, NTC, and 5V. Each cell in the table contains a radio button. Below the table is a green 'Update UO Status' button.

CHANNEL	DI	10V	NTC	5V
1	☐	☒	☐	☐
2	☐	☐	☒	☐
3	☒	☐	☐	☐
4	☒	☐	☐	☐
5	☒	☐	☐	☐
6	☒	☐	☐	☐
7	☒	☐	☐	☐
8	☒	☐	☐	☐

7. For example, to set Channel 1 of UI as NTC Sensor input, the radio button in front of Channel 1 must be selected as NTC and then “Update UI Status” button need to be pressed. This will make the Channel 1 of UI port as NTC sensor input port.

Update UO Status

CHANNEL	DO	AO
1	☐	☒
2	☐	☒
3	☐	☒
4	☐	☒
5	☐	☒
6	☐	☒
7	☐	☒
8	☐	☒

Note: Always restart the device after doing configuration for UI/UO Status, then only changes in configuration will take place.

8. The third table “UI_DATA” is for viewing the UI port data locally on the web page.

Data Section

UI_DATA	
CHANNEL	UI_DATA
1	421
2	117
3	0
4	0
5	0
6	0
7	0
8	0

9. User can select all the desired configuration at once and then press the Update button for UI or UO to update the Port status in the module.
10. User can change the IP and Mac Address for the TCP from the second tab “MODBUS TCP CONFIGURATION” in the web page and press the “Save Configuration” button and restart the device to set the new IP, as shown below.

UIUO PAGE	MODBUS TCP CONFIGURATION	UPDATE LOGIN
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Modbus TCP Configuration

MAC ID:
F8:B3:B7:A5:2F:2B

IP Address:
192.168.1.177

Port:
502

[Save Configuration](#)

Note: To reset the IP address to default IP, the user need to press and hold the reset button located near the Ethernet port for 5 seconds.

- On the third tab “UPDATE LOGIN” page, the user can change the Username and Password, to enhance the security of UIUO module, as show below.

UIUO PAGE	MODBUS TCP CONFIGURATION	UPDATE LOGIN
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Update Login Details

Current Username:
Enter current username

Current Password:
Enter current password

New Username:
Enter new username

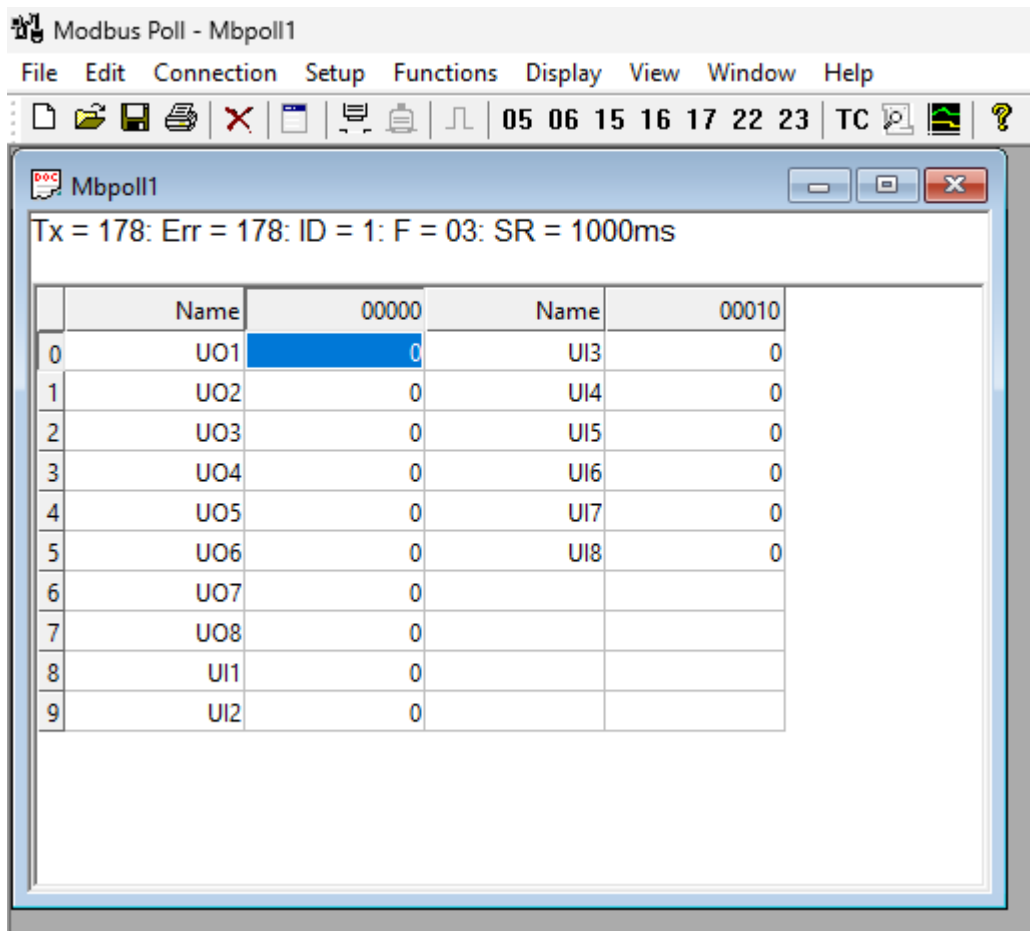
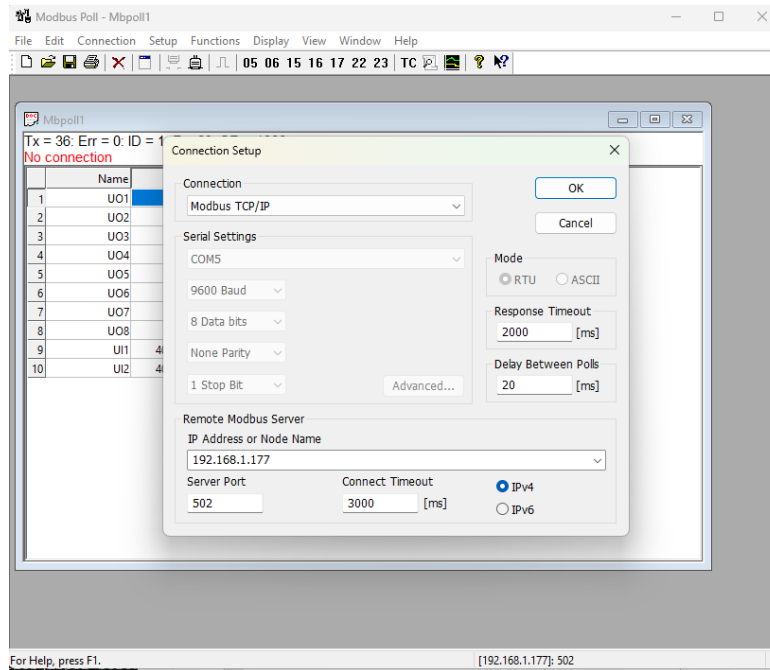
New Password:
Enter new password

Confirm New Password:
Re-enter new password

[Update Login Details](#)

- After all the configuration, User can now connect the UIUO module to Modbus TCP master at the set IP address or default IP address(192.168.1.177).

- Here we have shown a connecting with Modbus Poll software for demonstration in the below images.



14.The data of UI port can be fetched from the mentioned Modbus register in the table below, irrespective of DI, AI, NTC selection. The Desired Inputs are as follows:

15.

INPUT TYPE	Input Value
DI	0 V(0) , 12VDC (1)
AI(5V)	0 to 5VDC (0-1023)
AI(10V)	0 to 10VDC (0-1023)
NTC	NTC sensor(Temperature in °C)

16. Similarly, the data of the UO port can be set in the same register as mentioned in the table below. For DO, It should be sent 0 or 1 (12VDC). And for AO a 10-bit resolution value must be sent (i.e. from 0 to 1023) to set output of 0 to 10VDC.

OUTPUT Type	Output Value
DO	0 (0V) , 1 (12VDC)
AO	0 to 10VDC (0-1023)

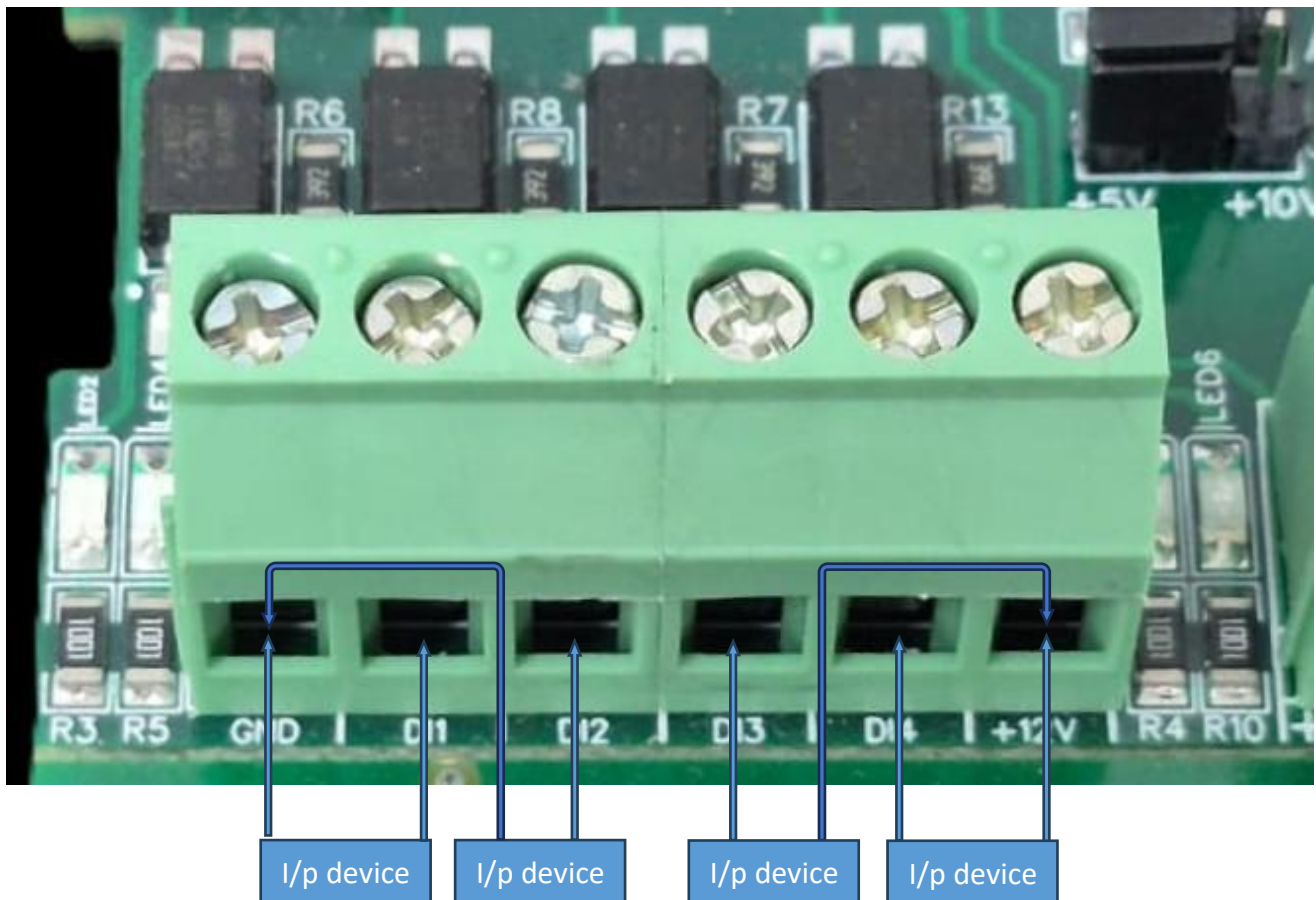
MODBUS COILS TABLE FOR UIUO PORTS:-

ID	Function Description	Register Description	Modbus Function Code	Data Type
1	UO 1 (DO)	00001	0x01	Boolean
2	UO 2 (DO)	00002	0x01	Boolean
3	UO 3 (DO)	00003	0x01	Boolean
4	UO 4 (DO)	00004	0x01	Boolean
5	UO 5 (DO)	00005	0x01	Boolean
6	UO 6 (DO)	00006	0x01	Boolean
7	UO 7 (DO)	00007	0x01	Boolean
8	UO 8 (DO)	00008	0x01	Boolean
9	UI 1 (DI)	00009	0x01	Boolean
10	UI 2 (DI)	00010	0x01	Boolean
11	UI 3 (DI)	00011	0x01	Boolean
12	UI 4 (DI)	00012	0x01	Boolean
13	UI 5 (DI)	00013	0x01	Boolean
14	UI 6 (DI)	00014	0x01	Boolean
15	UI 7 (DI)	00015	0x01	Boolean
16	UI 8 (DI)	00016	0x01	Boolean

MODBUS REGISTERS TABLE FOR UIUO PORTS:-

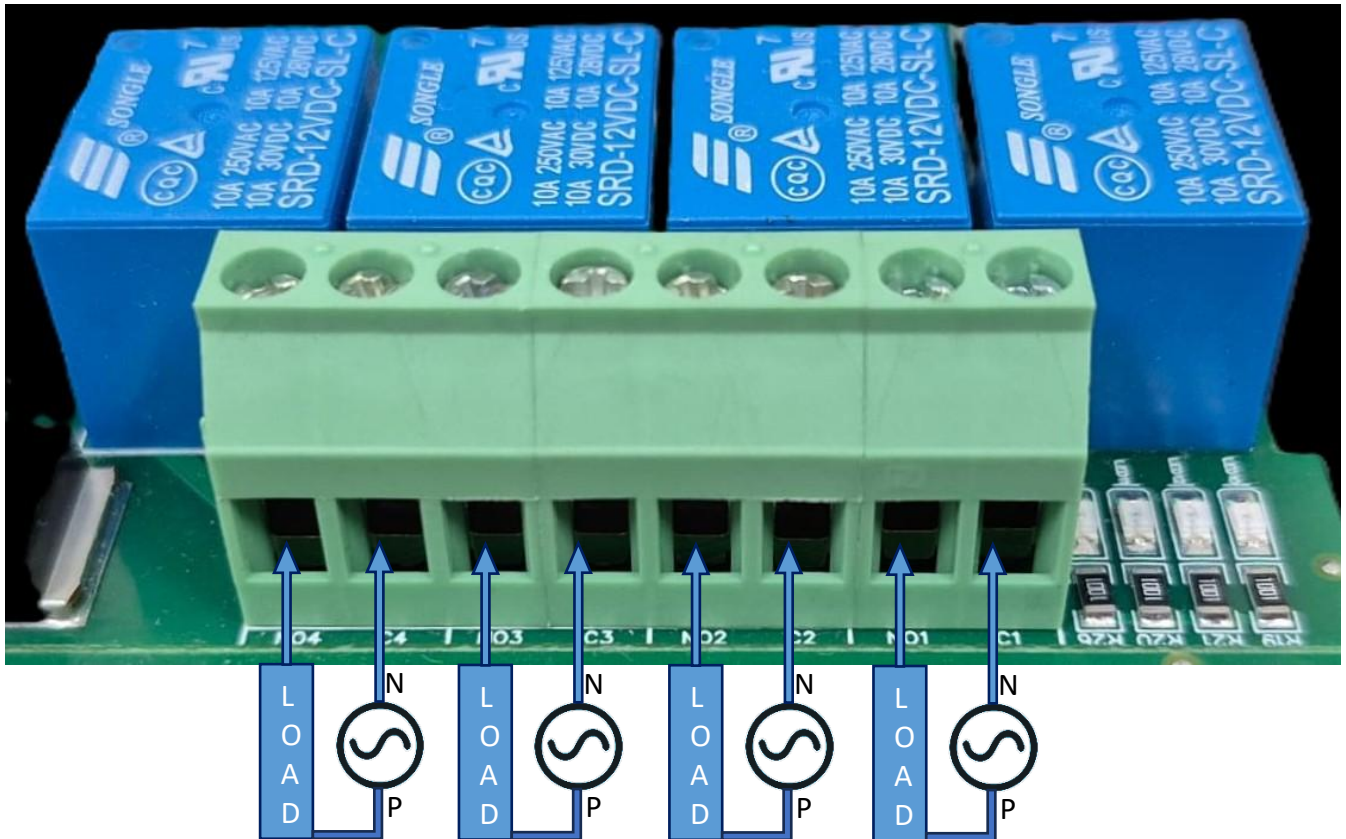
1	UO 1 (AO)	40001	0X03	16 Bit Unsigned int
2	UO 2 (AO)	40002	0X03	16 Bit Unsigned int
3	UO 3 (AO)	40003	0X03	16 Bit Unsigned int
4	UO 4 (AO)	40004	0X03	16 Bit Unsigned int
5	UO 5 (AO)	40005	0X03	16 Bit Unsigned int
6	UO 6 (AO)	40006	0X03	16 Bit Unsigned int
7	UO 7 (AO)	40007	0X03	16 Bit Unsigned int
9	UI 1 (AI,NTC)	40009	0x03	16 Bit Unsigned int
10	UI 2 (AI,NTC)	40010	0x03	16 Bit Unsigned int
11	UI 3 (AI,NTC)	40011	0x03	16 Bit Unsigned int
12	UI 4 (AI,NTC)	40012	0x03	16 Bit Unsigned int
13	UI 5 (AI,NTC)	40013	0X03	16 Bit Unsigned int
14	UI 6 (AI,NTC)	40014	0X03	16 Bit Unsigned int
15	UI 7 (AI,NTC)	40015	0X03	16 Bit Unsigned int
16	UI 8 (AI,NTC)	40016	0X03	16 Bit Unsigned int

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

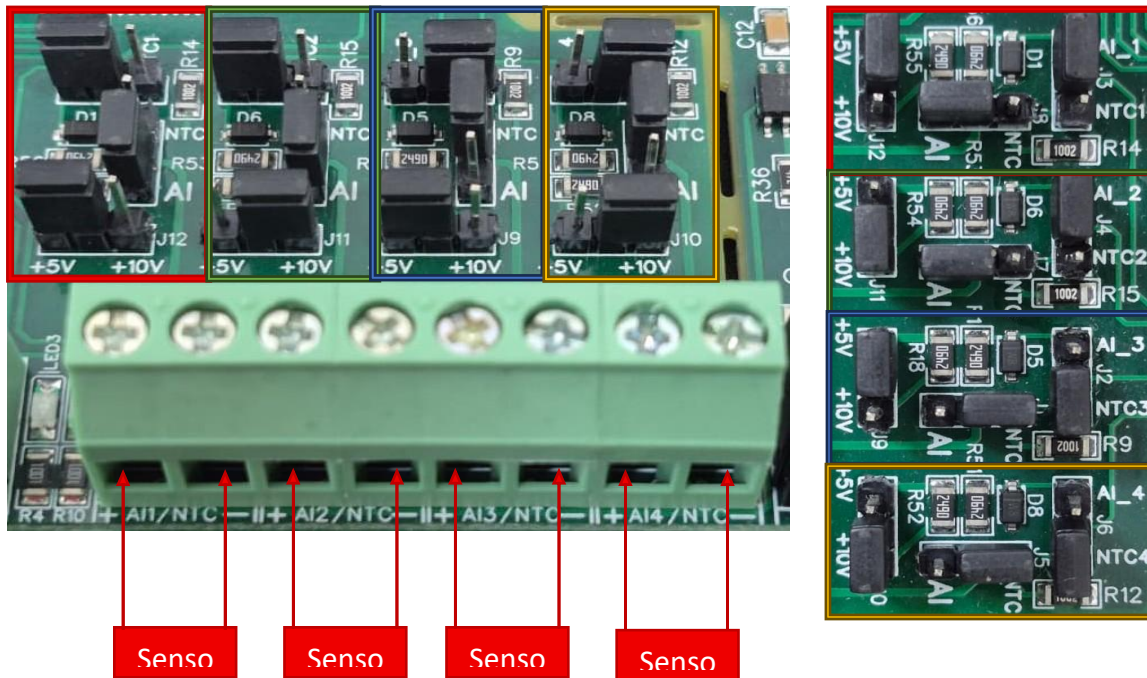
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

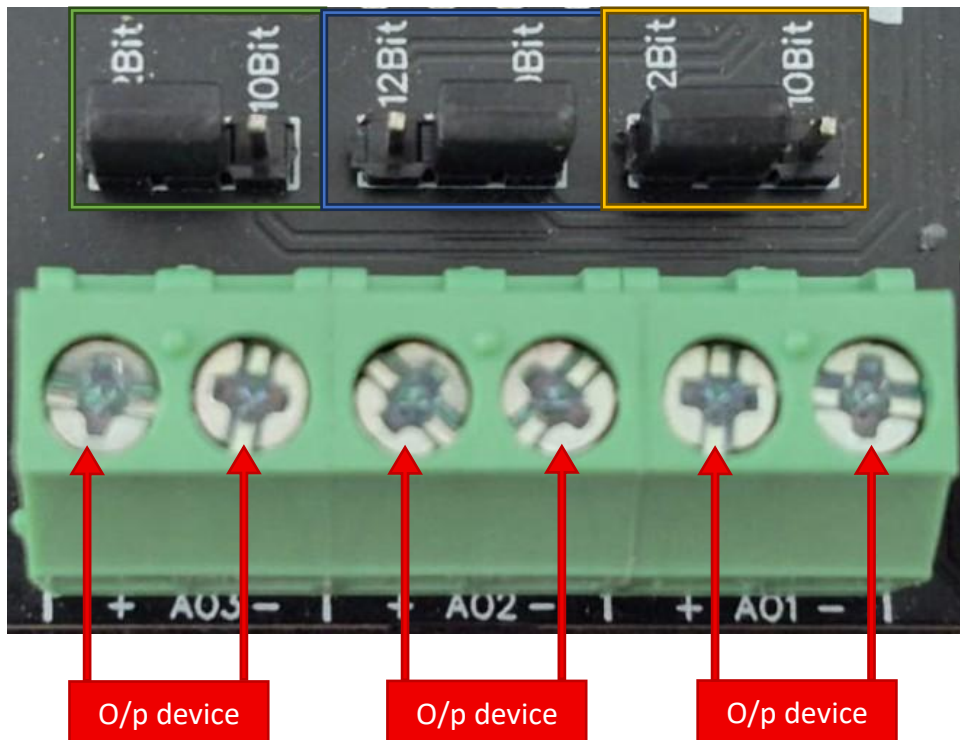
Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

AI-NTC

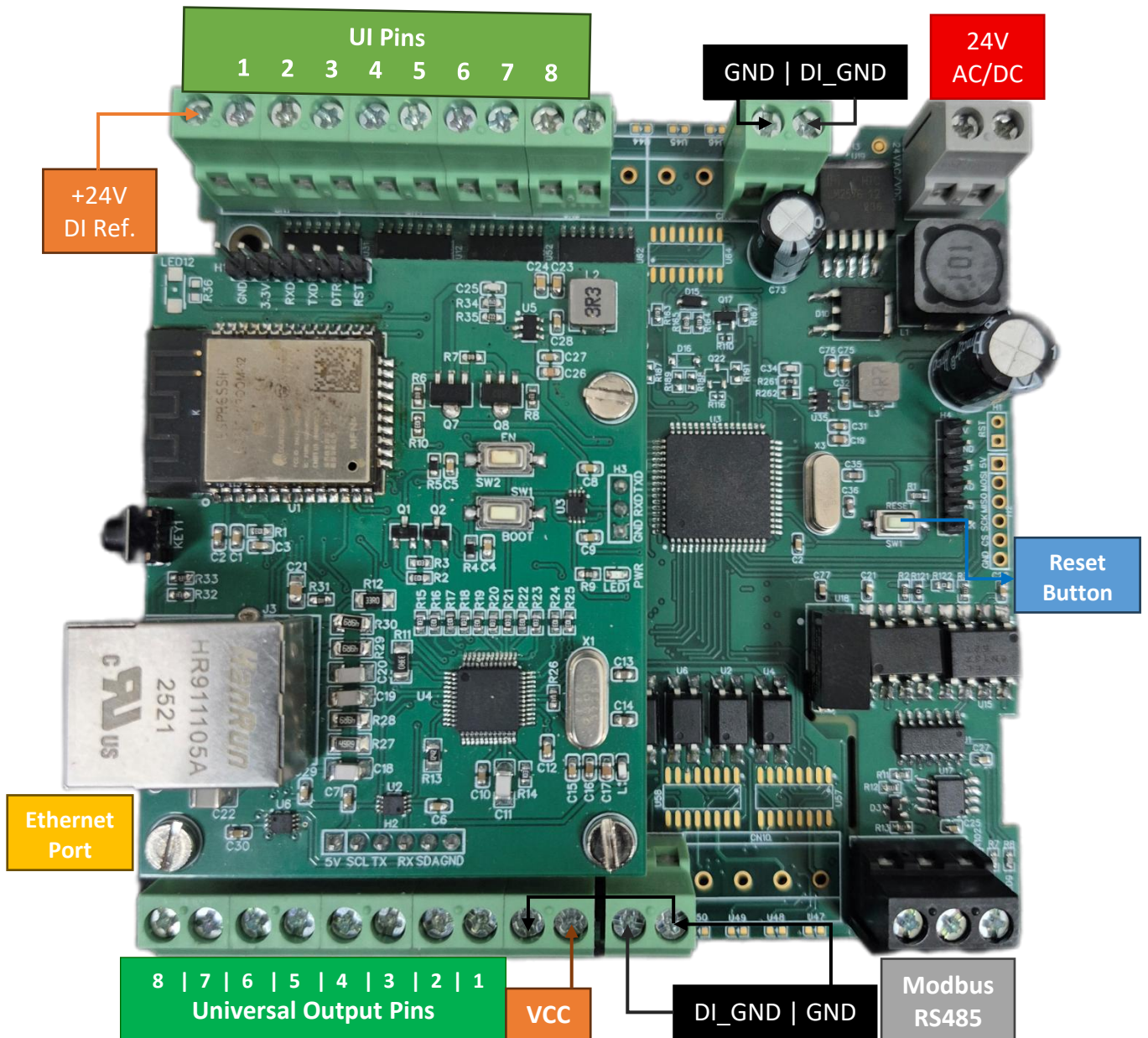


- **Analog Input & NTC Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to interface with either standard external analog sensors or NTC thermistors.
- **Channel Configuration (Visual Guide):** Each of the four channels features two dedicated jumpers to independently select the operating mode (AI or NTC) and the reference voltage range (5V or 10V). Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (AI Mode, 5V Range):** Set the mode jumper to **AI** and the voltage jumper to **+5V**.
 - **Green Box (AI Mode, 10V Range):** Set the mode jumper to **AI** and the voltage jumper to **+10V**.
 - **Blue Box (NTC Mode, 5V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+5V**.
 - **Yellow Box (NTC Mode, 10V Range):** Set the mode jumper to **NTC** and the voltage jumper to **+10V**.
- **Sensor Wiring Connections:**
 - The sensor or thermistor's positive signal wire should be connected to the + terminal of the desired channel (labelled AI1/NTC, AI2/NTC, etc.).
 - The sensor's common or ground wire should be connected to the corresponding - terminal.
- **Crucial Setup Recommendation:** Always ensure that both the mode jumper (NTC vs. AI) and the voltage jumper strictly match the hardware you are connecting. Incorrectly setting an NTC to an AI mode, or mismatching the voltage limits, will result in inaccurate temperature or signal readings.

AO



- **Analog Output (AO) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to send variable analog signals to **external output (O/p) devices**, such as variable frequency drives or proportional valves.
- **Channel Configuration (Visual Guide):** Each of the three AO channels features a **dedicated jumper** to set the **output bit resolution**. Refer to the color-coded boxes in the board diagram to configure your desired resolution:
 - **Green & Yellow Box (12-bit):** Set the resolution jumper to **12Bit**.
 - **Blue Box (10-bit):** Set the resolution jumper to **10Bit**.
- **Output Device Wiring Connections:**
 - The external device's positive signal wire should be connected to the + terminal of the desired channel (AO1, AO2, or AO3).
 - The device's common or ground wire should be connected to the corresponding - terminal..
- **Crucial Setup Recommendation:** Always ensure that the hardware jumper setting (10-bit or 12-bit) matches the output scaling configured in your main control software. A mismatch between software resolution and hardware configuration will result in incorrect output signal level



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